

2 Meiosis is a process that includes reduction division.

- (a) The cells that carry out meiosis in the male reproductive organ of plants are known as pollen mother cells.

Figure 2.1 shows photomicrographs of three different stages of meiosis in pollen mother cells from *Lilium*, a lily plant.

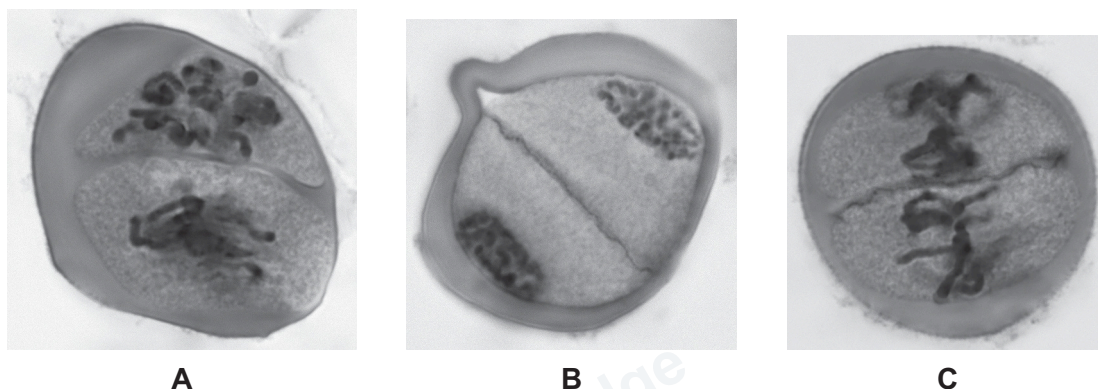


Figure 2.1

Name the stages of meiosis shown in cells **A**, **B** and **C** in Figure 2.1.

- A**
- B**
- C** [3]

- (b) Homologous chromosomes pair up and then separate into different daughter cells during meiosis.

Explain what is meant by a homologous pair of chromosomes.

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(c) Explain the need for reduction division to occur during meiosis.

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[Total: 9]